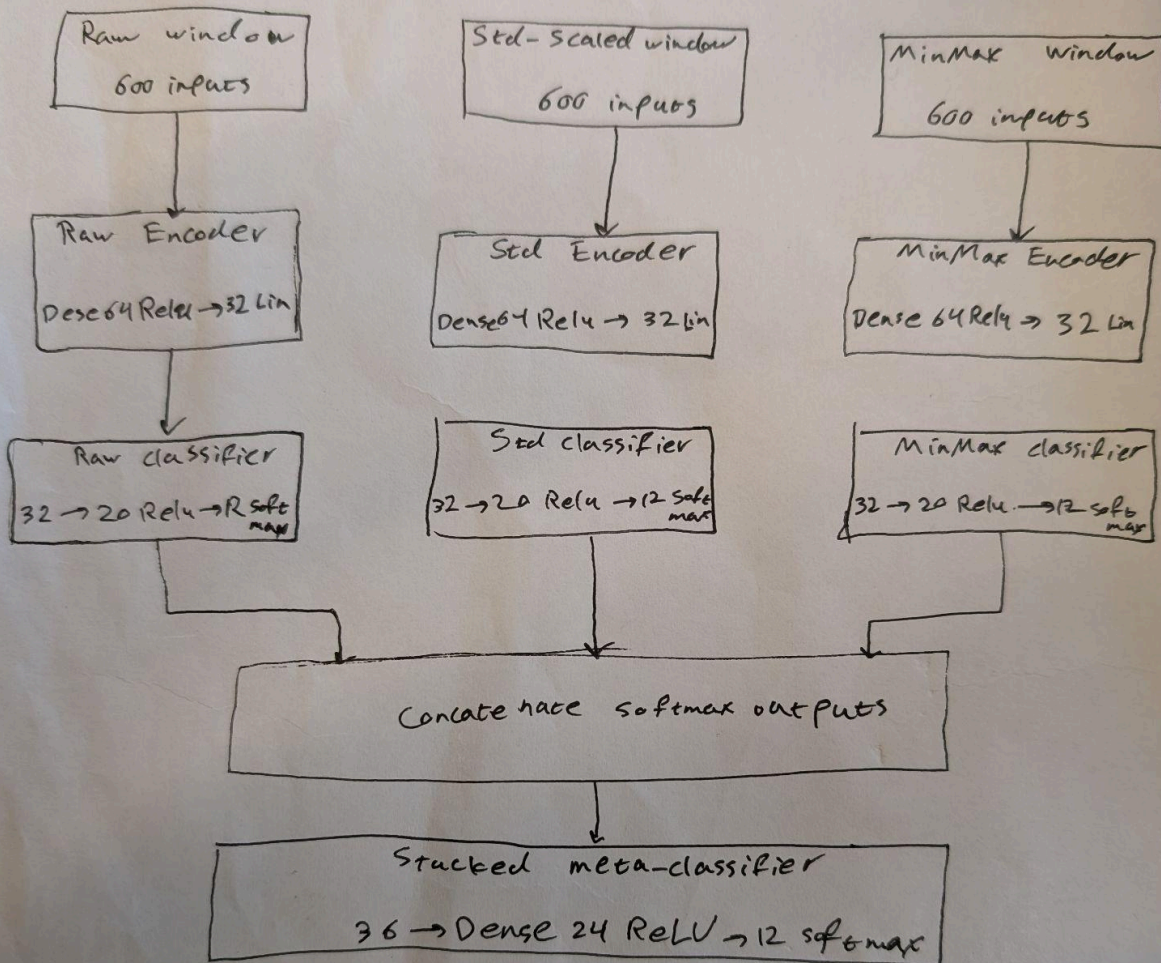


Part 1: <https://studio.edgeimpulse.com/public/1083283/live>

```
Sampling...
Timing: DSP 16 ms, inference 597 us, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  idle: 0.996094
  lift: 0.000000
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 21 ms, inference 597 us, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  idle: 0.996094
  lift: 0.000000
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 17 ms, inference 599 us, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  idle: 0.000000
  lift: 0.996094
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 21 ms, inference 598 us, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  idle: 0.000000
  lift: 0.996094
```

Part 2:

Q1:



Q2:

The models are first pruned from 10% to 80% sparsity, which gradually sets less important weights to zero and reduces the amount of useful model parameters. After pruning, the pruning wrappers are removed so the model can go through quantization aware training normally.

During QAT, the model is trained while simulating int8 quantization, so it can adjust to the loss in numerical precision before deployment. A mask is also used to keep the weights that were pruned at zero, so QAT does not accidentally make them nonzero again.

Finally, representative training samples are used during TFLite conversion to choose good ranges for int8 values. Overall, pruning reduces unnecessary weights, while quantization reduces the precision of the remaining weights and activations, making the model smaller and more practical for the Arduino.

Q3:

```
-----
Final prediction: Knees bending (crouching)
Class index: 7
Confidence score: 0.2773
-----
Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.8125, 0.0000, 0.0000, 0.0703, 0.1055, 0.0000, 0.0000, 0.0039, 0.0039
Standard-scaled model scores: 0.0000, 0.2930, 0.1680, 0.0000, 0.0625, 0.0000, 0.0000, 0.0000, 0.0000, 0.0625, 0.4102
Min-max-scaled model scores: 0.0000, 0.0000, 0.0195, 0.0000, 0.1836, 0.0000, 0.0156, 0.5937, 0.0000, 0.0000, 0.1836
Meta-classifier scores: 0.0195, 0.0039, 0.1914, 0.3359, 0.0156, 0.0117, 0.0273, 0.0625, 0.0000, 0.0000, 0.1797, 0.1484
-----
Final prediction: Walking
Class index: 3
Confidence score: 0.3359
-----
Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.8516, 0.0000, 0.0000, 0.1133, 0.0000, 0.0000, 0.0000, 0.0078, 0.0312
Standard-scaled model scores: 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.9961, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
Min-max-scaled model scores: 0.0000, 0.0000, 0.1719, 0.0000, 0.1562, 0.0000, 0.0195, 0.5000, 0.0000, 0.0000, 0.1562
Meta-classifier scores: 0.0039, 0.0039, 0.1562, 0.0664, 0.0000, 0.1094, 0.0039, 0.2930, 0.0000, 0.0000, 0.2148, 0.1484
-----
Final prediction: Knees bending (crouching)
Class index: 7
Confidence score: 0.2930
-----
Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.7461, 0.0000, 0.0000, 0.0000, 0.0977, 0.0000, 0.0000, 0.0039, 0.1484
Standard-scaled model scores: 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.9961, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
Min-max-scaled model scores: 0.0000, 0.0000, 0.1094, 0.0000, 0.0977, 0.0977, 0.2852, 0.3125, 0.0000, 0.0000, 0.0977
Meta-classifier scores: 0.0039, 0.0078, 0.2148, 0.0430, 0.0000, 0.2148, 0.0156, 0.1484, 0.0000, 0.0000, 0.2148, 0.1406
-----
Final prediction: Lying down
Class index: 2
Confidence score: 0.2148
-----
Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.5000, 0.0000, 0.0000, 0.0000, 0.5000, 0.0000
Standard-scaled model scores: 0.0000, 0.0195, 0.0078, 0.0000, 0.7969, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0781, 0.0977
Min-max-scaled model scores: 0.0000, 0.0000, 0.0117, 0.0000, 0.2695, 0.0000, 0.0039, 0.7148, 0.0000, 0.0000, 0.0000, 0.0000
Meta-classifier scores: 0.0000, 0.0000, 0.0117, 0.0586, 0.8867, 0.0078, 0.0000, 0.0078, 0.0000, 0.0000, 0.0273, 0.0000
-----
```

When testing the model on the Arduino, the predictions changed between classes such as walking, crouching, and lying down even when the motion was not always clearly matching those activities. The confidence was also fairly low for several predictions. This is likely because the model was trained on the mHealth dataset, while the Arduino has different sensor orientation, placement, and possibly different sensor scaling. The results could be improved by collecting training data directly from the Arduino and using prediction smoothing over multiple windows.

Q4:

The model often predicted knees bending (crouching) even when I was not actually crouching. This is probably because the model was trained on the mHealth dataset, while the Arduino has different sensor placement, orientation, and scaling. Because the live IMU data does not perfectly match the training data, some motions can look more like the crouching pattern to the model. Collecting training data directly from the Arduino would likely make the predictions more reliable.